

ROUGE-1, Worked Out by Hand

The classic word-overlap metric for generated text — it counts shared unigrams between a candidate and a reference, then balances precision and recall into an F1. Every count, division, and the harmonic mean shown explicitly.

What this document does. It takes one candidate sentence and one reference, counts their overlapping words (with the clipping rule that stops repeats from cheating), and computes ROUGE-1 precision, recall, and F1 by hand. It explains what each of the three numbers tells you and where overlap metrics break down. Numbers from Appendix A.

Context: ROUGE is a *reference-based* metric — it needs a gold answer to compare against. The reference-free, LLM-graded RAGAS metrics are PDFs #25–#28.

1 The idea: overlap with a reference

To score a generated summary or answer automatically, compare it to a human-written *reference* and measure how much they share. ROUGE-1 (“Recall-Oriented Understudy for Gisting Evaluation,” unigram version) uses the simplest unit of overlap: individual words. Two questions, two numbers:

- **Precision:** of the words the model *produced*, how many were in the reference? (Penalises padding.)
- **Recall:** of the words the reference *contains*, how many did the model produce? (Penalises omission.)

Intuition

ROUGE-1 treats both texts as bags of words and asks how much they overlap. Recall asks “did you cover the reference?” — crucial for summaries, where leaving out the key fact is the cardinal sin. Precision asks “did you avoid filler?” — a candidate that dumps every word would trivially have high recall but terrible precision. F1 forces the model to do both: say the right things *and* not much else.

2 The formulas

Let m be the number of overlapping unigrams (counted with *clipping*: a word matches at most as many times as it appears in the reference).

$$P = \frac{m}{|\text{candidate}|}, \quad R = \frac{m}{|\text{reference}|}, \quad F_1 = \frac{2PR}{P + R}$$

F_1 is the *harmonic* mean of P and R — it sits closer to the smaller of the two, so you cannot rescue a bad recall with a great precision (or vice versa); both must be decent.

3 Worked by hand

reference = the cat sat on the warm mat (7 words)
 candidate = a cat sat on the mat (6 words)

Step 1 — count overlapping unigrams (clipped).

word	in ref	in cand	clipped match
cat	1	1	1
sat	1	1	1
on	1	1	1
the	2	1	1
mat	1	1	1
a	0	1	0 (not in ref)

Overlap $m = 5$. (Note **the** appears twice in the reference but once in the candidate, so it contributes $\min(2, 1) = 1$ — clipping in action.)

Step 2 — precision, recall, F1.

$$P = \frac{5}{6} = 0.8333, \quad R = \frac{5}{7} = 0.7143,$$

$$F_1 = \frac{2(0.8333)(0.7143)}{0.8333 + 0.7143} = \frac{1.1905}{1.5476} = 0.7692.$$

The candidate is fairly precise (83% of its words are in the reference — only **a** is extra) but misses two reference words (**warm**, and one **the**), dragging recall to 71%. F1 lands at 0.77, nearer the weaker recall.

Mini-example · why the harmonic mean?

Suppose a lazy model outputs just the single word **cat**. Then $P = 1/1 = 1.0$ (perfect precision — its one word *is* in the reference) but $R = 1/7 = 0.14$. The arithmetic mean would flatter it at 0.57; the harmonic mean gives $F_1 = \frac{2(1)(0.14)}{1.14} = 0.25$, correctly damning it. F1 refuses to reward gaming one axis while ignoring the other — exactly what you want from a quality score.

4 What ROUGE misses

ROUGE-1 counts *surface words*, not meaning, so it has real blind spots:

- **Synonyms score zero.** “The feline rested” vs “the cat sat” shares almost no unigrams, yet means the same — ROUGE punishes valid paraphrase.
- **Word order is ignored.** A bag of words cannot tell “dog bites man” from “man bites dog.” (ROUGE-2 / ROUGE-L recover *some* order via bigrams / longest common subsequence.)
- **It needs a reference.** No gold answer, no ROUGE — useless for open-ended generation where many answers are valid.

These gaps are precisely why modern RAG evaluation leans on embedding-based and LLM-as-judge metrics (#25–#28): they score *semantics*, not token overlap. ROUGE remains useful as a cheap, deterministic, reproducible baseline — and for summarisation against fixed references, where word coverage genuinely matters.

Equation cheat-sheet

$$P = \frac{m}{|\text{cand}|}, \quad R = \frac{m}{|\text{ref}|}, \quad F_1 = \frac{2PR}{P + R} \quad (m = \text{clipped unigram overlap})$$

recall-oriented (coverage); blind to synonyms & order; needs a reference

Appendix A · Reference implementation

```
from collections import Counter
def rouge1(candidate, reference):
    c, r = Counter(candidate.split()), Counter(reference.split())
    m = sum(min(c[w], r[w]) for w in c)          # clipped overlap
    P, R = m / sum(c.values()), m / sum(r.values())
    F1 = 2 * P * R / (P + R) if P + R else 0.0
    return round(P, 4), round(R, 4), round(F1, 4)

rouge1("a cat sat on the mat", "the cat sat on the warm mat")
# (0.8333, 0.7143, 0.7692)
```

Internal learning material. Numbers reproducible from the appendix code. v1.0